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A wearable device for collecting multi-signal parameters of newborn

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ABSTRACT

In recent years, wearable health monitoring equipment has developed rapidly, but most functions are relatively single and can only monitor certain physiological parameters of the human body. This paper designed a wearable multi-physiological parameter monitoring system, which can continuously monitor multiple physiological parameters of newborns for a long time without affecting the normal human activities. The hardware circuit uses MSP430F1611 single-chip microcomputer as the central processing unit. The single-chip computer program completes the calculation of multiple parameters and sends them to the Bluetooth module through the serial port. The Bluetooth module receives the data and transmits it through the module antenna. After testing, each function module works normally. The results of the heart rate measurement experiments show that more than 96% of the heart rate data is within ± 3 BPM (Beat Per Minute). The experimental results of arterial oxygen saturation measurement show that the error of arterial oxygen saturation is within ± 2%. The results of arterial blood pressure measurement experiments show that the error between systolic and diastolic blood pressure meets the standard.

1. Introduction

Wearable medical devices are a major branch that gradually evolved with the development of wearable devices [1]. It mainly uses multimedia, wireless communication, and sensor technology to collect various physiological parameters of the human body to achieve monitoring of various physical signs of the human body [2]. Wearable devices have features such as wearability, mobility, and sustainability that can overcome these shortcomings of traditional medical devices. Its wearability and mobility enable users to use it at any time in any state of movement without going to the hospital for treatment, which not only reduces the user's own costs, but also reduces the hospital's precious resource occupancy rate [3–5]. The sustainability of wearable devices makes it possible to continuously record various physiological parameters of the human body for a long time, and these records are important for the diagnosis of some human diseases [6].

As a miniature electronic device that can be worn on the body to carry out activities, although scientific researchers have researched it as early as the 1960s and 1970s, it cannot be applied due to technical limitations [7]. However, thanks to the development of MEMS (Micro-electromechanical Systems) technology and other embedded low-power, miniaturized technologies, wearable devices have gradually appeared in the lives of the people, not only bringing unprecedented experience to users, but also driving the first development of scientific research [8]. Wearable devices are also the research hotspots of many scientific and academic institutions at home and abroad [9–11]. For example, the Parkinson shoe developed by the University of Delaware controls the vibration frequency of the insole through a built-in microcomputer to help patients with Parkinson's disease overcome the problem of difficulty in movement [12]. The My Heart system realizes the integrated design of ECG (Electrocardiograph) electrodes and wearable clothing, integrates ECG measurement module and acceleration measurement module into wearable clothing, and improves comfort in use [13]. Its main function is to monitor the changes of the ECG early to reduce the sudden risk of cardiovascular disease. The Code Blue system uses small-sized, low-power portable signal oximeters, lead electrocardiographs, and other sensors to measure physiological parameters such as blood oxygen saturation, heart rate, and electrocardiogram [14]. It uses wireless short-range technology to coordinate with mobile terminals. In this way, the data can be stored in the file and can be displayed in real time on the terminal, and an alarm will be issued if the physiological parameters exceed the normal range [15]. Relevant scholars have designed a medical monitoring system based on ZigBee wearable sensors [16–18]. The medical wearable sensors are used to collect the patient's physiological parameters, and the processed data is sent to the wireless communication module, and finally transmitted to the monitoring center by the ZigBee wireless network. Researchers have studied the relationship between blood pressure and pulse characteristic parameters, proposed a method for continuous measurement of blood pressure based on multiple pulse characteristic parameters,

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and established characteristic regression equations suitable for different people, but did not elaborate the correlation between diastolic blood pressure and pulse wave characteristic parameters [19,20].

This paper introduces the overall hardware design of the sign detection equipment. The hardware circuit is mainly composed of pulse wave acquisition module, ECG acquisition module, single-chip processor module, Bluetooth module and voltage management module, and each module is explained in detail. The software design includes ECG R-wave positioning, PPG (Photoplethysmography) signal first-order difference maximum point positioning, calculation of heart rate, blood oxygen saturation, and blood pressure, as well as the design of each module. We perform a functional test on the health parameter monitoring system for each module, compare the existing measurement equipment, perform a comparative experiment on each health parameter measurement of the system, analyze its errors, and verify the accuracy of the system.

The rest of this paper is organized as follows. Section 2 studies the principle and analysis method of neonatal multi-signal parameter collection, and in Section 3, the hardware design and software design of neonatal multi-signal parameter collection equipment are performed. Section 4 conducts system test experiments and analysis. Finally, the full text is summarized in Section 5.

2. Research on the principles and analysis methods of neonatal multi-signal parameters

2.1. Principles and detection methods of multiple physiological signals

(1) Heart rate pulse rate signal

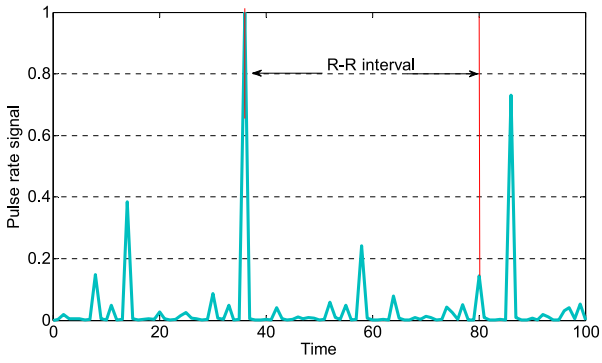
Heart rate refers to the number of beats per minute and pulse rate refers to the number of pulses per minute. Usually the heart rate and pulse rate are equal, but the pulse rate is not noticeable due to vascular sclerosis or other reasons and cannot be detected. The difference in heart rate and pulse rate between different individuals is quite obvious. The speed of heart rate and pulse rate is affected by many aspects such as gender, age, and mood. In a calm state, the heart rate of a normal person is generally between 60 and 100, the heart rate of a newborn is faster between 100 and 150, the heart rate of a child is between 70 and 130, and the heart rate of a professional athlete is between 40 and 60. For a normal person, a heart rate above 100 in a calm state is called tachycardia, and a heart rate below 60 is called bradycardia.

Both ECG and PPG are caused by heart beat, so heart rate and pulse rate can be obtained through ECG and PPG. We extract the R-value points of the electrocardiogram, determine an ECG cycle with two R-values to calculate the heart rate, and similarly extract the PPG trough points, and use the two trough points as a period to calculate the pulse rate. Fig. 1 shows the normalized pulse rate and heart rate signals.

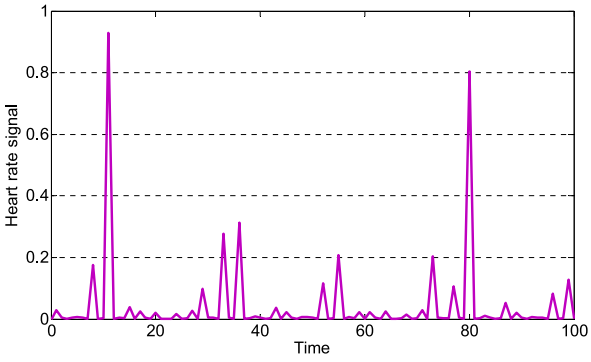
(2) Body temperature signal

Human body temperature refers to the average temperature in the deep part of the body. Body temperature is divided into deep temperature and surface temperature. Body temperature is also an important indicator to measure human health. At room temperature, the average temperature of a person's forehead skin is 33–34 °C, the body trunk temperature is 32 °C, and the hand temperature is usually 30 °C. The temperature of the human body changes slightly with changes in ambient temperature. This paper uses the partial pressure method to detect the human body surface temperature in real time.

The multi-physiological information detection system adopts a method of combining a thermistor and a fixed value resistor to obtain a change in temperature by measuring a change in voltage across the thermistor. Thermistors have various classification methods, usually divided according to the temperature coefficient. They are divided into positive temperature coefficient thermistors and negative temperature coefficient thermistors. Positive temperature coefficient and negative temperature coefficient refer to the resistance of the thermistor as a function of temperature. As the temperature increases, the thermistor



(a) Change trend of pulse rate signal after normalization



(b) Change trend of normalized heart rate signal

Fig. 1. Normalized pulse rate and heart rate signals.

increases and the temperature decreases. This multi-physiological information detection system uses the negative temperature coefficient thermistor series, which has the advantages of high sensitivity and wide measurement range. The relationship between the thermistor resistance and temperature is:

$$R_t = R \times EXP\left(B \times \left(\frac{1}{T_1} - \frac{1}{T_2}\right)\right) \tag{1}$$

In the formula, R is the nominal resistance value of the thermistor at normal temperature, B is the material constant of the thermistor, B of the thermistor used by this sensor is 4250 K, and the resistance value of the thermistor at normal temperature is 100 K.

(3) Blood pressure detection

According to the arterial hemodynamics, the pulse wave velocity is mainly related to the elasticity of the blood vessel wall and the thickness of the blood vessel. The relationship between them can be expressed by the Moens–Korteweg equation as follows:

$$v = \sqrt{\frac{gEa}{pd}} \tag{2}$$

In the formula, v is the pulse wave velocity, g is the acceleration of gravity, E is the elastic modulus of the blood vessel wall, p is the blood density, a is the thickness of the blood vessel wall, and d is the inner diameter of the blood vessel.

Because the elastic modulus E and the blood pressure P have an exponential relationship:

$$E = E_0 e^{\gamma P} \tag{3}$$

In the formula, E_0 refers to the elastic modulus when the pressure is 0, P refers to the blood pressure of the human body, and γ is a constant characterizing the characteristics of blood vessels. The value is 0.016–0.018 mm Hg.

Furthermore, the conduction time T of the pulse wave is inversely proportional to its conduction velocity v, which can be expressed as:

$$v = \frac{S}{T} \tag{4}$$

In the formula, S is the distance that the pulse wave travels.

2.2. Multi-signal parameter preprocessing

During the measurement process of ECG, PPG signals and respiratory signals, the real-time signals collected will drift due to human motion, power interference, etc., and the frequency of the baseline drift of the signals is low (generally less than 1 Hz). The methods used to remove the baseline drift of the signal include stationary wavelet transform and empirical mode decomposition. However, the wearable multi-physiological parameter detection system designed in this paper needs real-time signal processing inside the microcontroller. Wavelet transform and empirical mode decomposition have high requirements on processor computing performance and are not suitable for use in embedded systems. This paper uses comb filtering to remove baseline drift.

The comb filter is composed of a series of high-pass filters and low-pass filters. The comb filter arranges the passband or stopband at a certain interval to filter out noise and harmonics of a specific frequency. Because the amplitude–frequency response of the comb filter is a series of the same passband and stopband, similar to the teeth of a comb, it is called a comb filter. The comb filter is also called a delay filter. In principle, the comb filter uses the delay, multiplication, and superposition of the original signal to remove noise.

Because the breathing signal contains more high-frequency noise, and considering the effect of the filter on the original signal, a Butterworth low-pass filter is designed in this paper. The filter can be expressed as:

$$|H(w)|^2 = \frac{1}{1+\varepsilon^2\left(\frac{w}{w_P}\right)^{2n}} \tag{5}$$

Among them, w_p is the passband edge frequency.

ECG, PPG and breathing signals are mixed with a lot of noise signals. Among them, power frequency interference signals are more obvious in ECG. After considering various factors such as algorithm complexity and signal stability, this paper chooses a five-point moving average filter to filter out power frequency interference. The system equation of the five-point moving average filter is:

$$H(Z) = (1 + Z^{-1} + Z^{-2} + Z^{-3} + Z^{-4})/5 \tag{6}$$

The amplitude–frequency response curve is shown in Fig. 2. It can be seen that the filter has a good filtering effect on 50 Hz power frequency noise and its harmonic noise of each order.

2.3. Extraction of eigenvalue points of neonatal multiple sign parameters

The method of extracting physiological signals in this paper is called the differential threshold method. The differential threshold method has a high accuracy rate for periodic signals with obvious rising edges. The rising edges of breathing, ECG, and PPG signals are all obvious, but different signals are different Therefore, the algorithm needs to be adjusted accordingly.

Multi-physiological signals are discrete time series X (n), and they are first differentiated:

$$X'(n) = \frac{X(n) - X(n-1)}{\Delta n} \tag{7}$$

We square the part with derivative greater than 0 to highlight the rising part of the signal, as shown in the following formula:

$$\begin{cases} X'(n) \le 0 & X'(n) = 0 \\ X'(n) > 0 & X'(n) = (X'(n))^2 \end{cases} \tag{8}$$

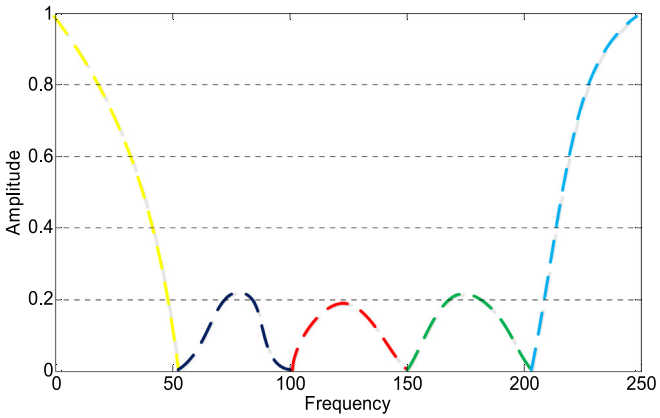


Fig. 2. Amplitude–frequency response curve of a five-point moving average filter.

We sum the processed signal windows to get the summed signal SSF (n):

$$SSF(n) = \sum_{K=0}^{W} X'(n-K) \tag{9}$$

The window width is appropriately adjusted according to different signal types, preferably the width close to the rising edge. If the window width is too narrow, it is easily affected by the glitch noise in the signal. If the window is too long, the rising part of the signal will not be obvious.

After the signal SSF (n) accumulated after the window is obtained, the position of the peak of the signal SSF (n) is between the peaks and valleys of the multi-physical signal, and the peak and valley points of the signal are obtained by locating the peak points of the SSF (n) signal. The window width W_1 takes 3–5 times the period of the signal waveform, as shown in the following formula:

$$SUM(n) = \sum_{K=0}^{W_1} SSF(n-K) \tag{10}$$

We set the threshold T of the extracted signal SSF (n), average the summed signals and then expand times, and the value of times is adjusted according to the steepness of the signal:

$$T = \frac{SUM(n)}{W_1} \times times \tag{11}$$

We filter the points in SSF (n) that are greater than the threshold value by threshold, compare SSF (n) with threshold value T, record the coordinates of the point when the value of SSF (n) is greater than threshold value T, and shield a distance S. Without comparison, S takes about two thirds of the signal period. And it dynamically adjusts with the change of the cycle:

$$\begin{cases} SSF(n) > T \\ n = n + S \end{cases} \tag{12}$$

We repeat the above steps to extract all characteristic value points of multiple physiological signals.

3. Design of equipment for collecting neonatal multiple signs parameters

3.1. Hardware design of multi-signal parameter acquisition equipment

(1) Overall hardware design

The hardware design of the device is mainly composed of pulse wave acquisition module, ECG acquisition module, single-chip processor module, Bluetooth transmission module and power management module. The overall hardware design diagram is shown in Fig. 3.

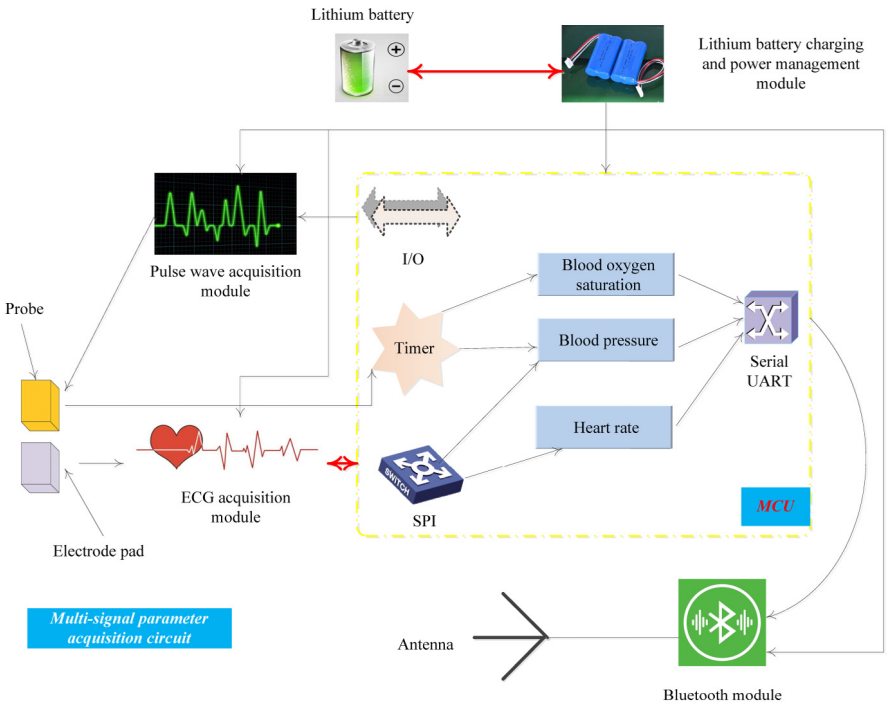


Fig. 3. Schematic diagram of the overall hardware design.

This design mainly collects the ECG signal and pulse wave signal, calculates the heart rate value from the ECG signal, calculates the blood oxygen saturation from the two sets of pulse wave signals according to Lambert–Beer law, and then uses the characteristic points of the pulse wave signal and the ECG. The time difference between the characteristic points is used to estimate the blood pressure value, so as to obtain the three vital sign parameters of the subject's heart rate, blood oxygen saturation, and blood pressure.

The two sets of sensors, ECG electrode pads and blood oxygen probes, collected the ECG signals of the cheeks on both sides and the pulse wave signals of the forehead. The ECG acquisition module amplifies the analog ECG signals and converts them into digital signals, and then sends them to the single-chip microcomputer processing module through the SPI (Serial Peripheral Interface) bus; the blood oxygen probe converts the reflected light intensity in the blood into frequency. The signal is then sent to Timer A of the microcontroller. The single-chip microcomputer processing module obtains three physical parameters of heart rate, blood oxygen saturation, and blood pressure through algorithm processing, and then sends them out by the Bluetooth module. The entire circuit is powered by a 3.7 V lithium battery. The power management module converts the 3.7 V voltage to 3.3 V and provides it to other modules.

(2) Blood oxygen acquisition module

In order to realize the miniaturization and comfort of the acquisition probe, an optical frequency converter and a two-color light LED (Light Emitting Diode) are embedded in a sponge pad, and the distance between the two is 1.8 cm. Because the distance between the detector and the light source is proportional to the depth of the measured tissue and inversely proportional to the intensity of the reflected light, and for PPG acquisition, it is mainly to detect the blood information of the arterioles in the muscle of the subcutaneous tissue.

The optical frequency converter has three pins: output, ground, and power. Its interior is composed of light intensity sensing devices, amplifiers, integrating circuits, etc., which directly converts the detected light intensity into corresponding frequency signals for output.

When the optical frequency converter works normally, its internal photodiode first obtains the light intensity information, converts the light intensity information into a current signal, and then uses the internal current–frequency converter to convert the current signal to a frequency.

(3) Pulse acquisition module

In practical applications, because the thickness of the human fingertip is relatively thin compared to other human tissues, and the light intensity detected by the photoelectric pulse sensor through the fingertip is relatively large, the measurement site is generally at the fingertip of the human body. According to the light receiving mode, the photoelectric pulse sensor can be divided into two types: transmissive and reflective.

The transmissive photoelectric pulse sensor is mainly composed of an infrared light source generating element and an infrared receiving element. The infrared receiving transistor has the characteristics of converting optical signals into electrical signals, and can generate electric energy under the irradiation of infrared rays. In order to accurately obtain the time relationship of the pulse, it is necessary to place the infrared receiving device and the infrared transmitting device at relative angles, and the light can pass through the fingers at a direct angle. In this way, the change in blood flow intensity by Microsoft can also cause the transmitted light intensity variety. At the same time, it can also eliminate the interference of the remaining light to the infrared light as much as possible.

(4) ECG acquisition module

The ECG signal is collected using a differential method of two electrodes, and the lead method is a bipolar single lead. The ECG signal enters 2 channels. After amplification, A/D conversion (sampling rate is 500 Hz), becomes a 3-byte ECG digital signal. The ECG signal of the second channel is three bytes of zero on the first channel (the first channel is not used, the data is zero). The first three bytes of the frame constitute a digital signal with a total of 9 bytes per frame. Once ADS1292 has a frame of data conversion completed, it sends an interrupt signal to the microcontroller through the DRDY pin, and then the data is sent to the microcontroller processing module through the SPI interface. The chip uses an external active crystal oscillator with an oscillation frequency of 2.048 MHz.

(5) Signal conditioning module

The pulse signal output by the low-pass filter amplifier circuit is then entered into the U2 and U3 op amps to perform waveform

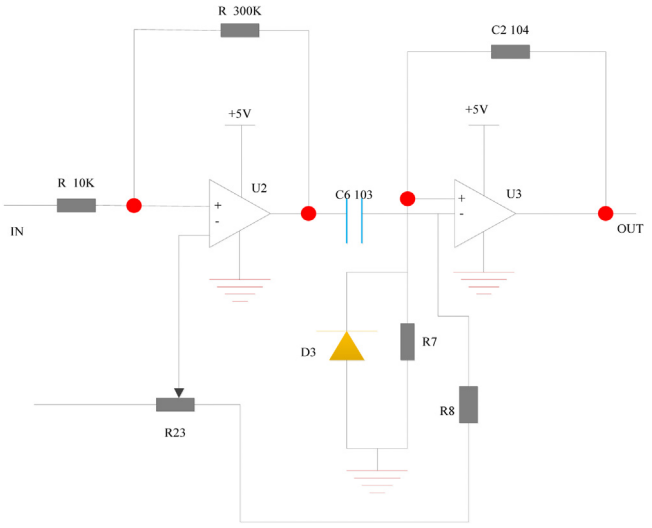


Fig. 4. Waveform shaping circuit.

shaping. The circuit is shown in Fig. 4. U2 is a voltage comparator, C6 and R8 form a differentiator, and U3, C6 and R7 form a monostable multivibrator with a pulse width.

U2 will output a high level at each trailing edge of the comparator input signal, causing R7 to charge C6. The charging current of C6 gradually decreases. When the potential of the non-inverting input terminal of U3 drops below the potential of the inverting input terminal, it will output a low level again, and then repeat. The pulse signal is output from the output end of U2 and connected to the PIO pin of the Bluetooth module. The microprocessor in the Bluetooth chip detects the rising and falling edges of the level to count the number of pulses and calculate the heart rate.

(6) MCU processing module

The microcontroller control module uses an ultra-low-power consumption microcontroller MSP430F1611, and the current in the operation mode is 330 μA (below 1 MHz, 2.2 V). This microcontroller is a 16-bit ultra-low-power microcontroller with fast operation speed. It integrates ultra-low frequency oscillator, 12-bit analog-to-digital converter, 16-bit timer, UART asynchronous communication serial port, analog comparator, etc.

The communication between the microcontroller and the Bluetooth module is through the serial TXD1 and RXD1 two lines, and the calculated physical parameters are sent to the Bluetooth module. The baud rate used is 115 200 bps. The microcontroller uses an external clock crystal oscillator with an oscillation frequency of 8 MHz.

(7) Bluetooth module

The Bluetooth module uses the chip CC2540, which supports three low-power modes for easy low-power design. When the chip receives the data sent by the microcontroller through the serial port, it immediately forwards the data to the antenna sending port, and then sends the data out through the external ceramic antenna. The transmitting and receiving distance of the antenna can reach 10 m, which meets the design requirements.

(8) Charging and power management module

This design is a wearable device, which requires a lithium battery to power the device. The battery uses a 3.7 V lithium battery at 1000 m Ah. The charging chip uses CN3052A, and its charging current can be adjusted by resistor R3. In this design, the charging current is set to 160 m A. The charging interface uses a micro USB interface, which is convenient to use a USB cable to charge the battery. When the external power source is charging the battery, the LED indicator D1 is on, and when it is not charging or the battery is fully charged, D1 is off. The voltage stabilization chip TLV70033 converts the battery voltage of 3.7 V to 3.3 V, and supplies power to the circuit of the entire device.

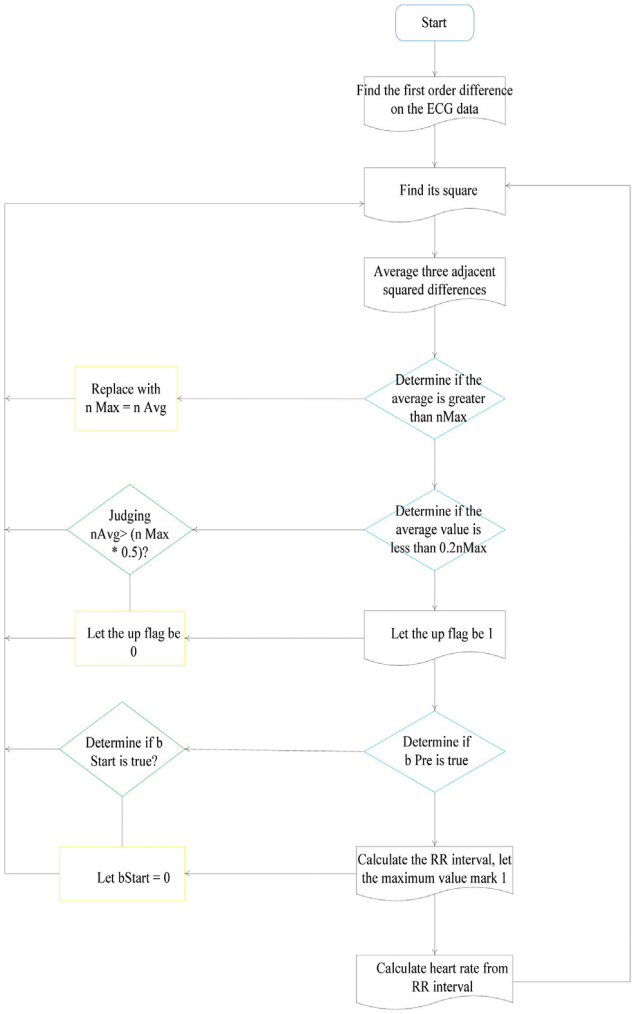


Fig. 5. Flowchart of ECG R-wave positioning and heart rate calculation.

3.2. Software design of multi-signal parameter acquisition equipment

(1) ECG R-wave positioning and calculation of heart rate value

To determine the real-time heart rate value, the time of a periodic ECG signal must be determined. The ECG starts from the P wave. The PP interval represents the time of a periodic ECG signal. Generally, there is no difference between the PP interval and the RR interval. R waves are taller and easier to locate, so RR intervals are often used in engineering to calculate heart rate.

The ECG R-wave positioning uses the first-order difference square threshold method, and the program flowchart is shown in Fig. 5. When n Avg is less than 0.3 times n Max, it can be considered that n Avg goes downward from the maximum point, and the uplink flag b Pre is set to 1. At this time, if (! B Start && b Pre) is true, it is considered that the maximum point (R wave position) is found, and the interval from the previous maximum position is obtained. If the interval is greater than 40, it can be determined as the maximum point. Finally, the heart rate value is obtained from the interval.

(2) Calculation of PPG signal components

The conventional method for obtaining the DC and AC components of a PPG signal is to obtain the PPG by locating the maximum and minimum points of the PPG signal and using interpolation to supplement the data between the maximum and minimum points. The upper and lower envelopes of the signal. The difference between the upper and lower envelopes is the AC component, and the upper envelope is the DC component.

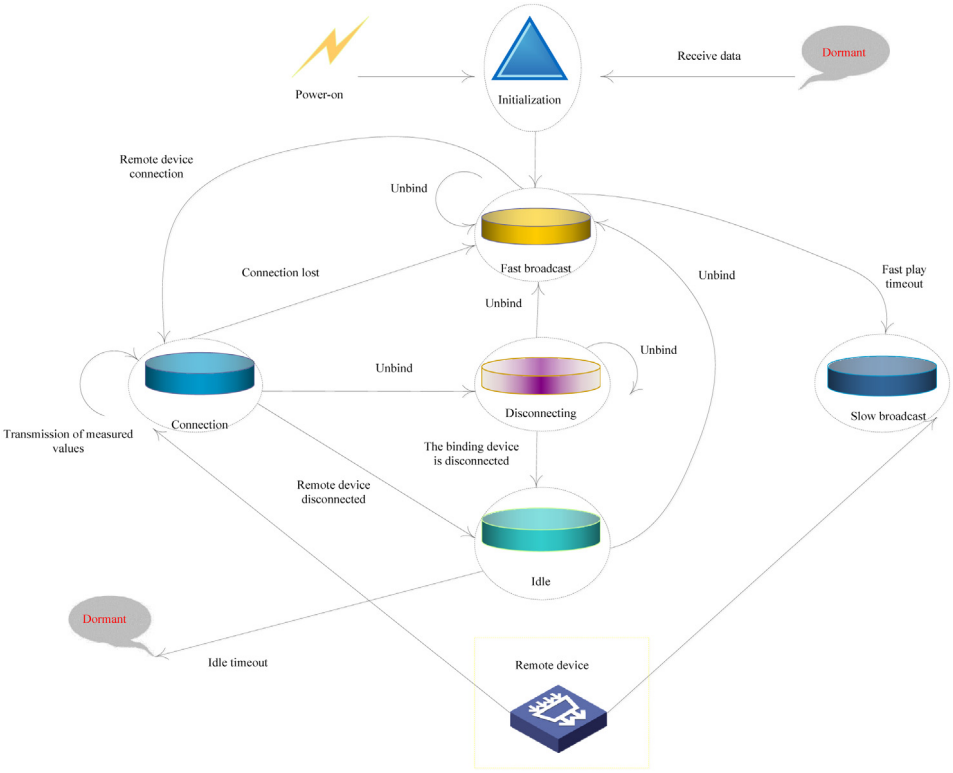


Fig. 6. Bluetooth internal state machine.

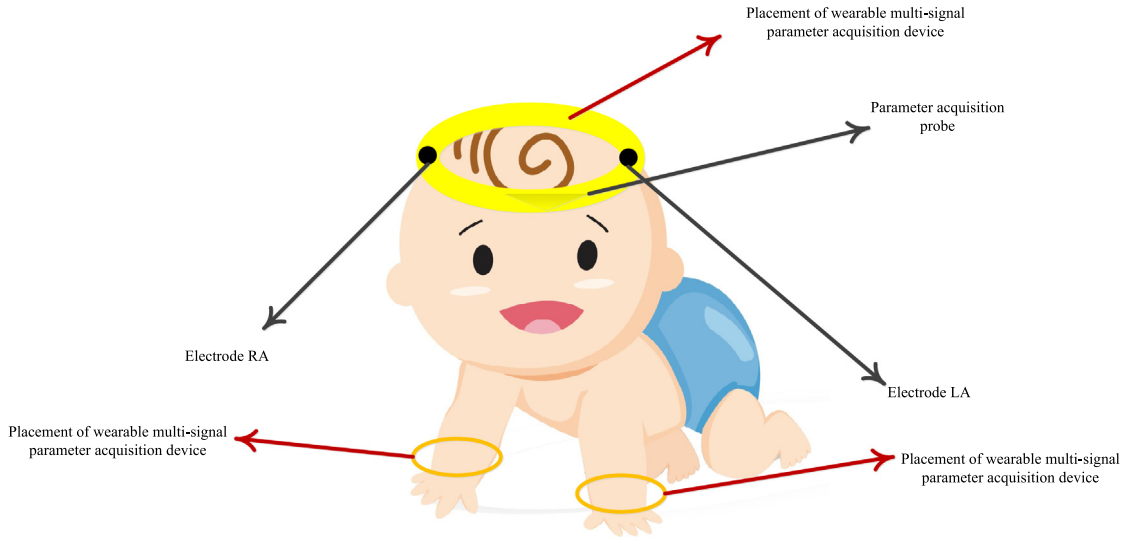


Fig. 7. Schematic diagram of the overall design.

In practice, in order to reduce the amount of calculation, the amplitude difference between the maximum point and the minimum point of the PPG signal is approximated as an AC component, and the average value of the PPG signal is approximated as a DC component.

(3) Design of Bluetooth low energy connection and communication

The program has six internal states, and the state machine is shown in Fig. 6.

(1) Initialization state: When the program is powered on or reset, the system will enter the initialization state, which mainly completes the registration of services and prepares for entering the fast broadcast state.

(2) Disconnected state: In this state, the collector waits for confirmation of disconnection.

(3) Idle state: In this state, the Bluetooth pulse signal collector is not connected to any device.

(4) Connection state: In this state, the Bluetooth pulse signal collector has established a connection with the mobile device, and the Bluetooth collector sends heart rate data.

(5) Fast broadcast status: It sends broadcast information at a faster frequency so that it can be discovered and connected.

(6) Slow broadcast state: This state can only be entered through the fast broadcast state. Broadcast information has been sent at a slower frequency, mainly to save energy consumption.

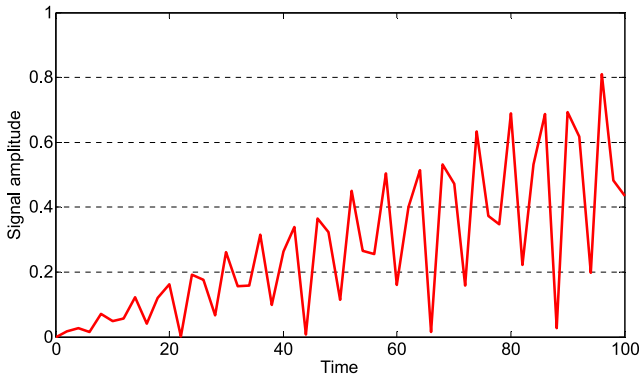


Fig. 8. Signal waveform.

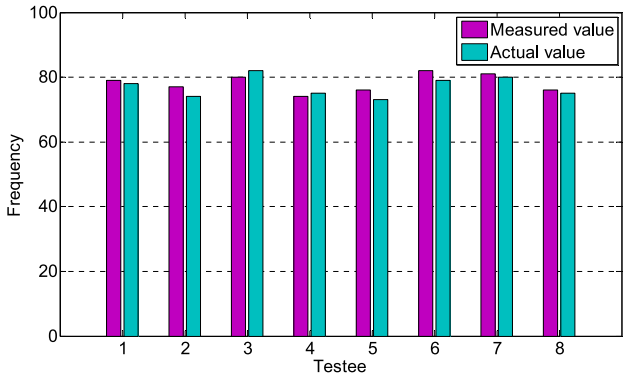


Fig. 9. Pulse signal acquisition results of 8 different test objects.

4. System test experiment and analysis

4.1. Wear scheme design

Due to the limitation of newborn conditions, and considering the convenience of wearing, this solution places all sensors and hardware devices in the range of the newborn's head and wrist. The program design is shown in Fig. 7.

Two conductive silicone patches are placed on both sides of the jaw band as electrodes to collect the ECG signals on both sides. ECG signal acquisition was performed using a bipolar single-lead method. The two conductive silicone sheets are connected to the interface of the acquisition device through two thin wires. A black elastic band is wrapped around the head as a headband, and the oximetry probe placed on the headband is closely attached to the hairless forehead. The blood oxygen probe is composed of a bi-color light LED and an optical frequency converter, and is used to collect the bi-color light PPG signal at the forehead. The blood oxygen probe is connected to the interface of the acquisition device through five thin wires. The physical sign information acquisition device is placed in the space on the top of the helmet and is used to process the electrical signals sent by the ECG electrodes and the blood oxygen probe, and send the calculated physical sign information through the Bluetooth module on the device.

The overall solution requires that the sensor be convenient to wear, the device be lightweight, and suitable for wearing on the head, without causing physical and psychological burden on the newborn.

4.2. Test experiment

(1) Signal waveform test

Before starting the test, it needs to check whether the circuit is soldered or short-circuited. After the test is OK, we use a voltage regulator to power the acquisition module (no Bluetooth module) to make the infrared sensor and signal conditioning module operate normally. It means that the power is normal and can work normally. This test only tests whether the signal acquisition and signal conditioning functions of the infrared sensor are working properly. We test the signal waveform, connect the DO output of the signal conditioning module to the oscilloscope, and observe the waveform after the signal stabilizes, as shown in Fig. 8.

Because the signal has been converted into a digital signal, it does not need to go through the ADC conversion circuit to identify the level amplitude and is directly detected by the Bluetooth module.

The experiment mainly selected 8 healthy newborns to participate in the test, and mainly counted the number of red lights flashing within one minute after the signal was stable for each newborn. The experimental results are shown in Fig. 9.

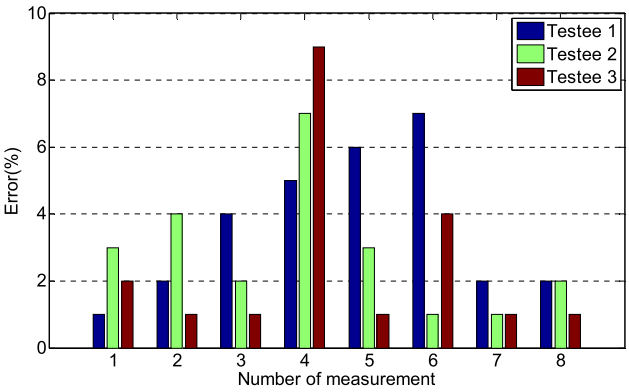


Fig. 10. Heart rate experiment error.

The experimental results show that the transmission pulse photoelectric sensor has a small error between the pulse number and the actual pulse number, which meets the design requirements.

(2) Parameter test

We test the accuracy of this system by comparing existing healthy wearable devices. For heart rate and blood oxygen, this paper selects the Yuyue brand model YX301 finger-type pulse oximeter for comparison; for arterial blood pressure, this paper uses the electronic pulse wave sphygmomanometer BP-37B produced by RAYCOME company for comparison.

In order to test the accuracy of the instrument, for different individuals, 3 testers were selected here to perform heart rate measurement experiments. Measurements were taken at rest, and each tester measured 8 times. We place the device at the designated detection position on the head, and place the Yuyue YX301 oximeter on the index finger of the left hand of the human body. The experimental error is shown in Fig. 10. It can be seen that the experimental results with an error greater than 8% only appeared on the third tester during the fourth test. There were no errors greater than 10% throughout the experiment.

For different individuals, 3 testees were selected here to perform arterial oximetry experiments. Measurements were taken at rest and each tester measured 8 times. We place the device at the designated detection position on the head, and place the Yuyue YX301 oximeter on the index finger of the left hand of a human body.

From the comparison data of arterial oxygen saturation measurement in Fig. 11, it is concluded that the single-point arterial oxygen saturation error of this design and comparison device is within ± 2%.

The method of measuring blood pressure by using pulse wave transit time is affected by individual differences, so the blood pressure accuracy test is performed on individuals who have established a model here. We use this device to collect the blood pressure of the human body and measure 8 times continuously. At the same time, we use the

Table 1
Blood pressure test results.

Serial number	Pulse transit time	Calculated systolic blood pressure	Absolute systolic pressure error	Calculated diastolic blood pressure	Absolute diastolic pressure error
1	20.11	120	6	87	10
2	20.43	121	5	82	7
3	20.38	120	4	86	8
4	20.45	119	3	78	6
5	22.12	118	4	79	9
6	24.31	123	5	72	8
7	28.87	107	3	73	11
8	28.19	107	1	80	10

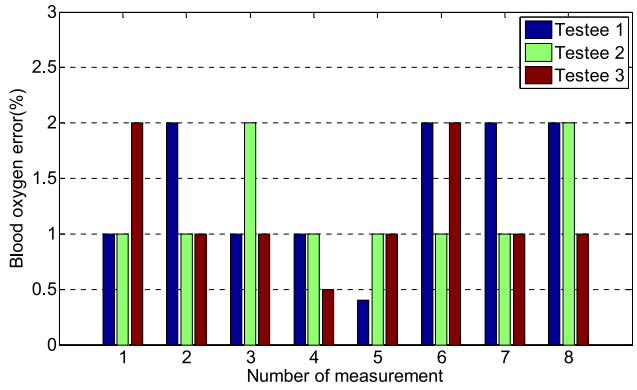


Fig. 11. Blood oxygen experiment error.

pulse wave electronic blood pressure monitor to measure the blood pressure of the right hand. The experimental results are shown in Table 1.

From Table 1, the maximum value of the absolute value of the single-point systolic pressure error of this design and comparison device is 6, and the maximum value of the absolute value of diastolic pressure error is 11.

(3) Error analysis

When measuring heart rate, the individual's psychological state at different times and changes in body position during the measurement will affect the final test results and increase errors.

In the blood oxygen empirical parameter calibration experiment, the experimental individuals are healthy people, and the blood oxygen saturation is within the normal range. Therefore, in the high blood oxygen saturation interval, the measurement error is small, which is close to the measurement value of the comparison device.

In the modeling of systolic and diastolic blood pressure and pulse wave transit time, each time the same individual is measured, the tightness of the band will have a certain effect on the pulse wave. If the bandage is too tight, the blood flow will be hindered for a long time, which will affect the calculation of the pulse wave transmission time, and then bring errors to the final test result.

5. Conclusion

This device includes a hardware circuit board and a blood oxygen collection headband. The hardware circuit uses the MSP430F1611 single-chip microcomputer as the central processing unit. The single-chip microcomputer captures the signals sent by the optical frequency converter and obtains two kinds of PPG signals of two-color light. The single-chip computer program calculates the three parameters of heart rate, blood oxygen saturation, and blood pressure, and sends it to the Bluetooth module through the serial port. The Bluetooth module receives the data and transmits it through the module antenna. The hardware circuit is powered by a 1000 m Ah lithium battery, and the lithium battery can be charged through the circuit's charging module. The blood oxygen collection headband is a black elastic band. The headband is wrapped around the upper side of the head by the two side hooks. A two-color light LED and an optical frequency converter are fixed in the middle of the headband. As a blood oxygen probe, it collects pulse wave signals. The system function test and the accuracy test of each parameter were completed, and comparative experiments were performed on each health parameter. The preliminary experiments show that the heart rate measurement error is within ± 4BPM (Beat Per Minute), and the error of more than 96% of the heart rate data is within ± 3BPM. Preliminary experiments show the feasibility of the system in the field of health multi-parameter monitoring. In the state of motion, the ECG and PPG signals will produce sharp changes and EMG interference, which will cause the wrong positioning of the signal and the detection of the parameters will cause a larger error. The next step is to study the algorithm to remove motion artifacts, so that the sign collection device can remove motion interference, so as to effectively monitor the signs of newborns in motion.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Xiqiu Hu: Data curation. **Jingrong Cao:** Formal analysis. **Hao Wu:** Validation.

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